TOPIC 2: UNDERSTANDING DATA TYPES AND PRE-PROCESSING

MACHINE LEARNING (SAIA 2113)

Session 2025/2026 – Sem 1

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COURSE LEARNING OUTCOMES

At the end of this topic, students should be able to:

a. Discover types of data utilized in machine learning.

b. Identify steps to perform data preprocessing.

c. Identify main challenges related to data in machine learning.

Topic 2: Understanding Data Types and Pre-Processing

- Discover types of data utilized in machine learning.
- Identify steps to perform data preprocessing.
- Identify main challenges related to data in machine learning.

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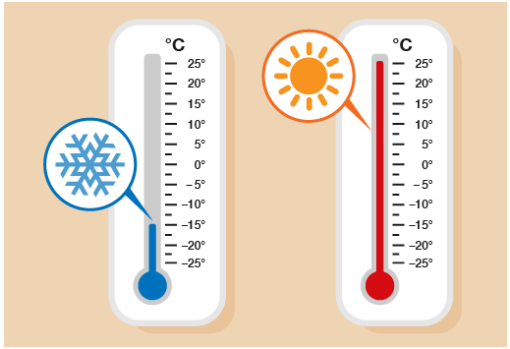
What is data?

- Data is a collection of facts, numbers, words, observations or other useful information.
- Artificial intelligence (AI) has further increased the demand for data, as machine learning models rely on it for accuracy.
- As data's volume and complexity grow, effective data management is crucial, alongside rising concerns for data security, privacy, and regulatory compliance.

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Data is core to AI algorithms

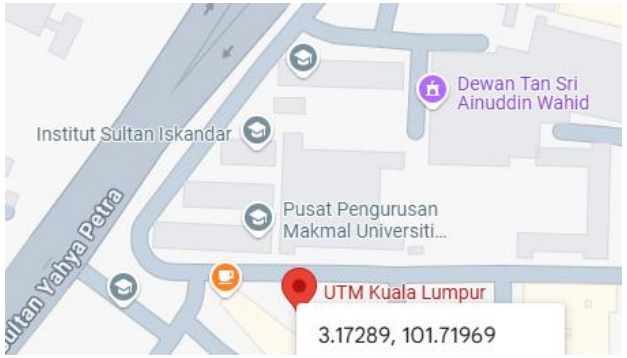
- Data is the new fuel and AI is the accelerator.
- Garbage in, garbage out (GIGO) emphasizes that the quality of a system's output is directly influenced by the quality of the input it receives.
- Incorrect choice of data, poorly represented data, or missing data, will cause AI algorithms will perform poorly.



Temperature



Customer reviews



Location coordinates

Types of data

- Data exists in various forms, each with distinct characteristics, sources, and formats.
- Different types of data serve different purposes.
- Additionally, a single data point or set may belong to multiple categories, such as structured and quantitative, or unstructured and qualitative.

Most common types of data include:

- **Quantitative data**
- **Qualitative data**
- **Structured data**
- **Unstructured data**
- **Semi-structured data**
- **Metadata**
- **Big data**

Quantitative data

Quantitative data is a type of data that can be expressed in numerical values.

- Height or weight of a person or object. (170 cm, 5’7”, 1.75 m, 70 kg, 154 lbs)
- Room temperature. (25 Celsius, 77 Fahrenheit)
- Scores and marks. (59, 80, 60)
- Time. (2.15 PM, 14:15 PM, 10 minutes)

Quantitative data are further classified into two parts: discrete data & continuous data.

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Discrete data

The discrete data contain the values that fall under integers or whole numbers.

- Total numbers of students present in a class. (30 students, 45 students)
- Cost of a cell phone. (RM 4999)
- Numbers of employees in a company. (50 employees, 500 employees)
- The total number of players who participated in a competition. (32 players, 100 players)
- Days in a week. (7 days)

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Continuous data

Continuous data represents information that can be divided into smaller levels(any value within a given range).

- Height of a person. (170.5cm, 181.3cm)
- Speed of a vehicle. (60.1 mph, 101.1 km/h)
- “Time taken” to finish the work. (2 hours, 30 minutes)
- Wi-Fi frequency. (2.4Ghz, 5Ghz)
- Market share price. (RM250.75, RM120.50)

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Qualitative data

Qualitative is a types of data that cannot be measured or counted in the form of numbers.

- Genders. (male, female)
- Languages. (English, Spanish, French, Mandarin)
- Favourite holiday destination. (Paris, Langkawi, Bali)
- Opinion on something. (agree, disagree, neutral)
- Colours. (red, blue, green, yellow)

Qualitative data are further classified into two parts: nominal data & ordinal data.

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Nominal data

Nominal data don’t have any meaningful order, their values are distributed into distinct categories.

- Colour of hair. (Blonde, Red, Brown, Black, etc.)
- Marital status. (Single, Widowed, Married)
- Nationality. (Indian, German, American)
- Gender. (Male, Female, Others)
- Eye Colour. (Black, Brown, etc)

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Ordinal data

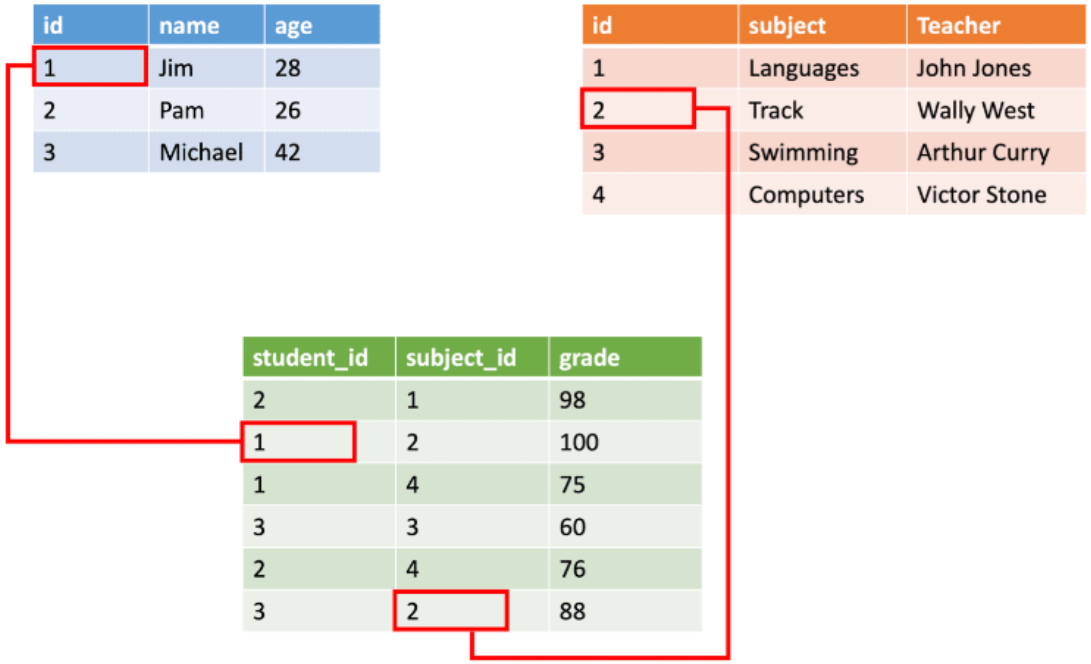
Ordinal data represents data that has a meaningful order among its categories.

- Feedback, experience, or satisfaction on a scale of 1 to 10.
- Letter grades in the exam. (A, B, C, D, etc.)
- Ranking of people in a competition. (First, Second, Third, etc.)
- Economic status. (Low, Medium, High)
- Education Level. (Primary, Secondary, Higher)

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Structured data

- Structured data is **systematically arranged in a well-defined format**, commonly found in relational databases or spreadsheets.
- Structured data are often organized into **rows and columns**.
- It may include both numerical information (prices, revenue figures) and descriptive data (categorical labels like “yes” or “no”, dates, names, addresses, credit card numbers).
- Examples of structured data: customer records and financial reports.



id	name	age
1	Jim	28
2	Pam	26
3	Michael	42

id	subject	Teacher
1	Languages	John Jones
2	Track	Wally West
3	Swimming	Arthur Curry
4	Computers	Victor Stone

student_id	subject_id	grade
2	1	98
1	2	100
1	4	75
3	3	60
2	4	76
3	2	88

Source: k21academy.com

Unstructured data

- Unstructured data lacks a fixed format.
- Unstructured data can include both **qualitative information (such as customer comments) and quantitative elements (numerical values embedded in text)**
- Examples (**textual data**): **Emails, text documents, social media posts, call transcripts, message texts.**
- Examples (**nontextual data**): **Image files, multimedia files, video files, mobile activity, sensor data from IoT devices.**



PICKS FROM THE EXPERTS

TREND 1

CONFLICT AND GEOPOLITICS

In recent years, the world has become more dangerous as war has broken out in the Middle East and on Europe's doorstep in Ukraine, while tensions remain high in the Taiwan Strait. According to the think tank Institute for Economics & Peace, there are currently 56 armed conflicts around the globe, the highest number since World War II. Meanwhile, numerous reports show a year-over-year double-digit increase in cyberattacks, as bandits and national armies seek to steal data and wreak havoc.

These trends are worrisome but also present an opportunity for investors. The conflicts around the globe have produced a shortage of weapons, particularly missiles and missile defense, making companies that cater to militaries a good bet for 2025.

Ron Epstein, a senior analyst at Bank of America, says the U.S. is expected to spend around $1 trillion on defense next year, and that he has buy ratings on all the major companies in the sector. But he singled out **RTX (ticker RTX)** as occupying a particularly strong position thanks to its Raytheon division's "fantastic missile franchise" and because its military business is complemented by a successful commercial aircraft unit.

Those with qualms about investing in weapons may prefer to put their money into the cybersecurity sector instead. There are a host of publicly traded companies that tout their ability to protect networks and thwart cyberattacks, including the growing number of attacks generated by AI. But performance across the sector has been uneven, as underscored by CrowdStrike, which issued a faulty update in July that grounded Delta's fleet of planes and cost the airline and other clients billions of dollars.

One cyber company on an upswing is **Fortinet (FTNT)**, based in Sunnyvale, Calif., with well over $5 billion in annual revenue. Like others in the sector, it does not issue dividends and has a fairly high P/E ratio. At the same time, Fortinet's share price has handily outperformed the broader market and most of its peers. Founded in 2000, the company is a trusted name in the industry and offers a wide variety of services, catering to a range of sectors such as finance, health care, education, and government.

Some investors who foresee global turmoil roiling markets may prefer to avoid individual company stocks altogether, and bet instead on volatility and uncertainty itself. While there are ETFs that loosely track the so-called VIX fear gauge, such plays are for sophisticated, active investors and don't belong in a buy-and-hold portfolio. A better option for those betting on ongoing turmoil is old-fashioned gold—specifically, through an ETF like **SPDR Gold Shares (GLD)** that permits you to hold the yellow metal at a reasonable cost. While GLD prices sank in November after the election of Donald Trump, many expect the ETF to continue its recent strong run. Goldman Sachs predicts gold will reach more than $3,000 an ounce in 2025, as conflict and trouble keep the world on edge.
—*J.J.R.*

TREND 2

A RETURN TO LUXURY SPENDING

This year has not been kind to the luxury sector. The share prices of upscale companies like Burberry Group, Kering, LVMH Moët Hennessy Louis Vuitton, and Movado are all down year to date. This is the result of Americans tightening their purse

MANY FLAVORS OF BULL Stocks have had a fantastic year in 2024, but Bitcoin did even better. The cryptocurrency got a still bigger boost from the election of Donald Trump, while bond prices took a hit.

YEAR-TO-DATE TOTAL RETURN FOR SELECTED INVESTMENTS
AS OF NOV. 12, 2024

Source: Fortune magazine

Semi-structured data

- Semi-structured data **combines elements of structured and unstructured data**.
- It **does not follow a strict format**, but it often contains tags or markers that simplify organization and analysis.
- Examples: XML files and JSON objects.

Example: JSON document

The following example displays the names of three guests in JSON.

```
{"guests":[
 { "firstName":"John", "lastName":"Doe" },
 { "firstName":"María", "lastName":"García" },
 { "firstName":"Nikki", "lastName":"Wolf" }
]}
```

Example: XML document

The following example displays the names of three guests in XML.

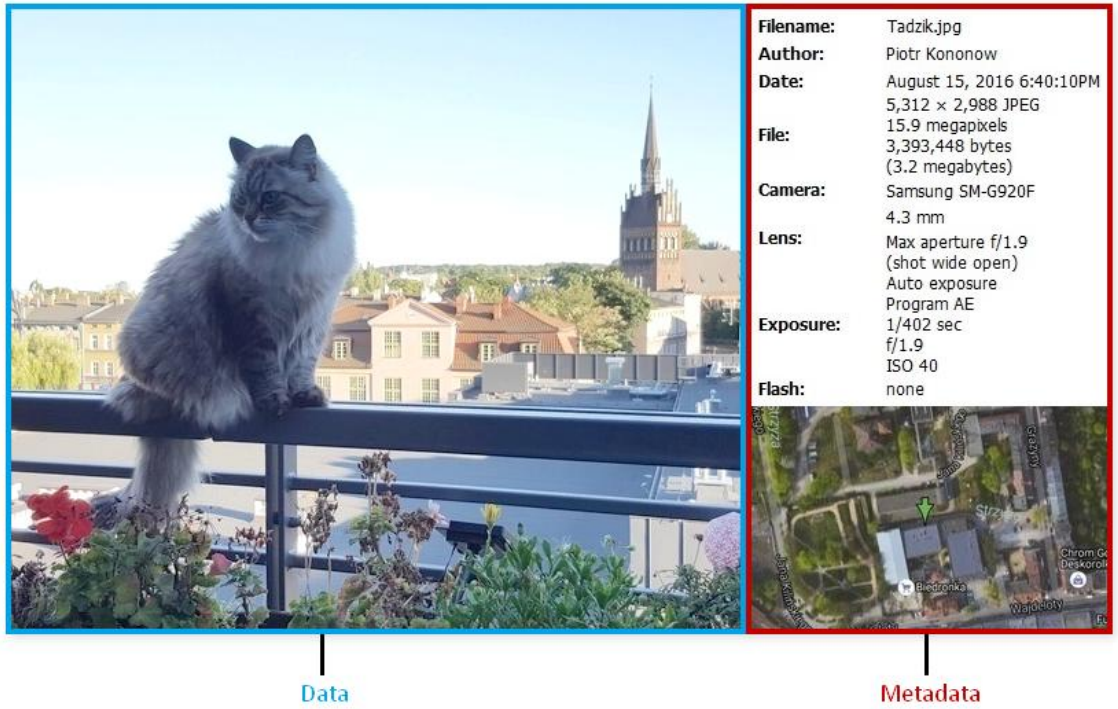
```
<guests>
 <guest>
  <firstName>John</firstName> <lastName>Doe</lastName>
 </guest>
```

Source: Amazon AWS

Metadata

- Metadata is essentially **information about other data**. It includes details about the characteristics of a data point or dataset, such as file names, authors, creation dates, or data type.

- Metadata **enhances data organization, searchability, and management**.



Source: dataedo.com

Big data

- **Big data describes massive, varied datasets that include structured, unstructured, and semi-structured information, all of which continue to expand rapidly.**

- Traditional data management systems cannot store, process, and analyze them because these datasets are so huge and complex in volume, velocity, and variety.

- Examples: Streaming services, social media, personalized medicine, autonomous driving.

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Where to get data?

Here are different ways to obtain data:

- Use public datasets. (Kaggle, Google Dataset Search, UCI Machine Learning Repository, data.gov.my)
- Collect your own data. (web scraping, manual data collection, IoT sensors)
- Use APIs. (Twitter API, Google Maps API, Open Weather API, Financial Data API)
- Generate synthetic data. (GAN - Generative Adversarial Networks)

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Dataset

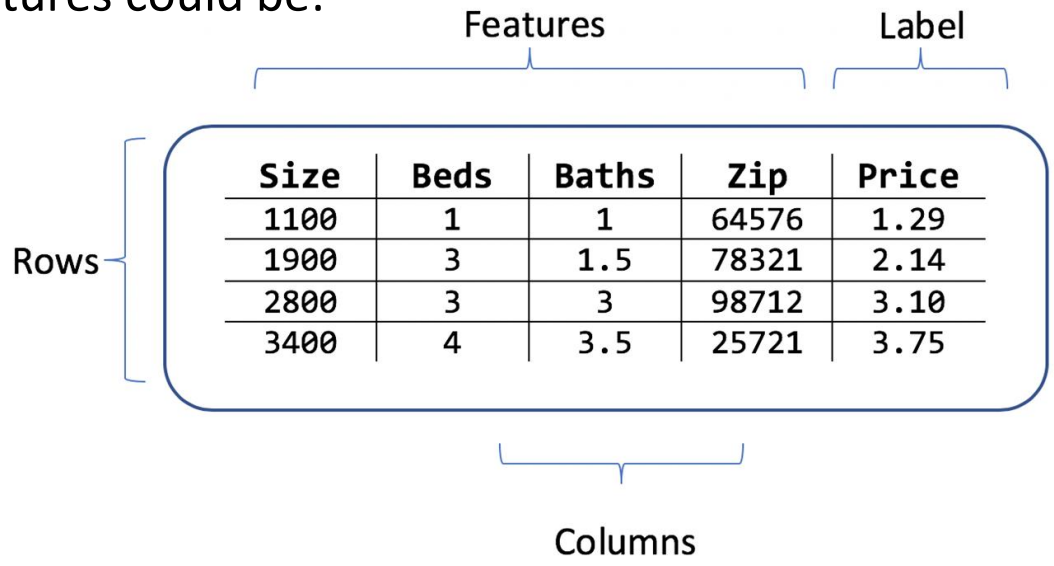
A dataset is a table of rows of training examples, and columns of attributes. Each row has:

- A label – the attribute that you want to predict.
- Features – the attribute that are used as inputs to predict the label.

For the house-price prediction scenario, the features could be:

- The square footage.
- The number of bedrooms and bathrooms.
- The zip code.

The label is the historical house price.



Source: Microsoft

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Topic 2: Understanding Data Types and Pre-Processing

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Data Preprocessing

Processing applied to raw data to prepare it for further analysis or processing tasks. Data preprocessing involves several steps:

- Data cleaning
- Data integration
- Data transformation
- Data reduction

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Data Cleaning

Data cleaning is the process of **identifying** and **correcting errors or consistencies** in the data.

For example:

- **Handling missing values** – Mean/Mode imputation, fill in or remove missing data.
- **Removing duplicates** – Eliminating duplicate records to ensure each entry is unique and relevant.
- **Correcting inconsistent formats** – Standardizing formats to maintain consistency.

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Data Integration

Data integration involves **combination data from multiple sources** to create a unified dataset.

For example:

- **Schema matching** – Aligning **fields and data structures from different sources** to ensure consistency.

- **Data deduplication** – Identifying and removing duplicate entries across multiple datasets.

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Data Transformation

Data transformation **converts data into formats suitable for analysis**, machine learning, or mining.

For example:

- **Scaling and normalization** – Adjusting numeric values to a common scale.
- **Encoding categorical variables** – **Converting categorical data into numerical values** using one-hot or label encoding techniques.
- **Feature engineering and extraction** – Creating new features or selecting important ones to improve model performance.

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Data Reduction

Data reduction **simplifies the dataset** by **reducing the number of features** or records while **preserving the essential information**.

Techniques for data reduction:

- **Feature selection** – Choosing the **most important features** contributing to the model's performance.
- **Principal component analysis (PCA)** – A dimensionality reduction technique that transforms data into a lower-dimensional space.
- **Sampling methods** – **Reducing the size of the dataset** by selecting representative samples is useful for handling large datasets.

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Exploratory Data Analysis

Exploratory data analysis (EDA) is used by **data scientists to analyse and investigate data sets and summarize main characteristics**.

Three main types of EDA:

- Univariate EDA - looking at a **single variable** at a time.
- Bivariate EDA - looking at **two variables** at a time.
- Multivariate EDA involves looking at three or more variables at a time.

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Exploratory Data Analysis

Graphical methods consists of **creating charts, graphs, and other visualizations** to explore the data. **Non-graphical methods** involve using **statistical techniques** to explore the data.

- Univariate non-graphical: The data analysed consists of just **one variable**. Since it's a single variable, it doesn't deal with causes or relationships. (**mean, median, spread, shape**)

- Univariate graphical: Non-graphical methods don't provide a full picture of the data. Graphical methods are needed.

- Multivariate non-graphical: Relationship between more than two variables of the data through **cross-tabulation or statistics**.

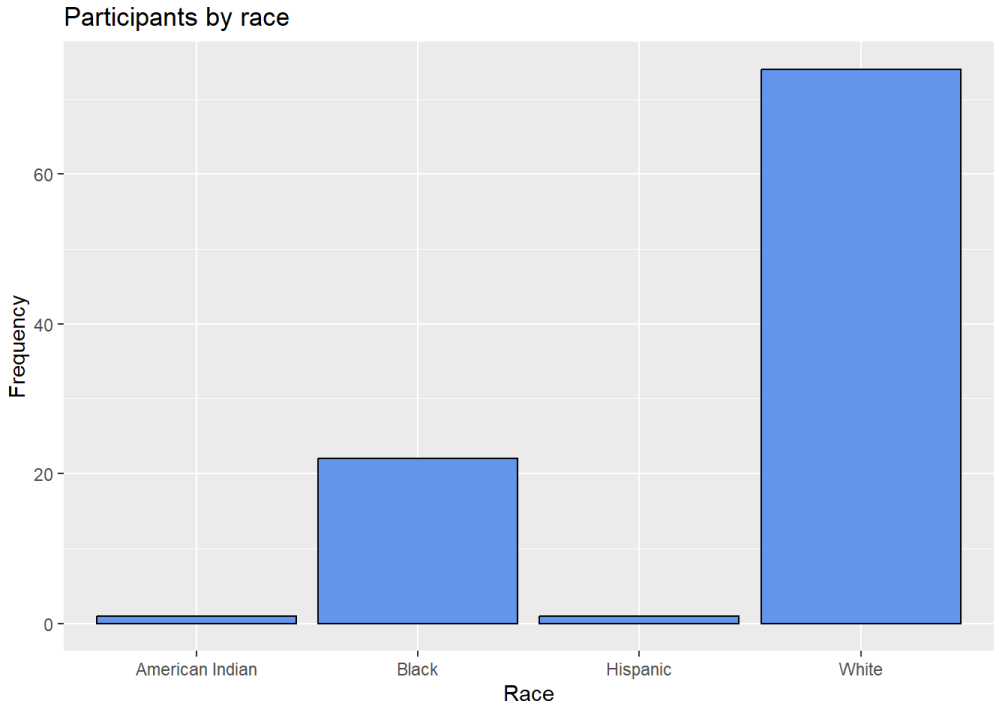
- Multivariate graphical: Uses **graphics** to display relationships between more than two variables of data.

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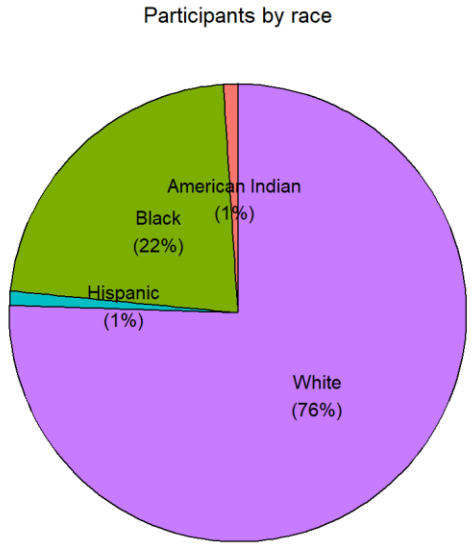
Exploratory Data Analysis

Univariate graphs for categorical variable:-

A bar chart is used to display the distribution of participants by race.



Pie chart is used to compare each category.

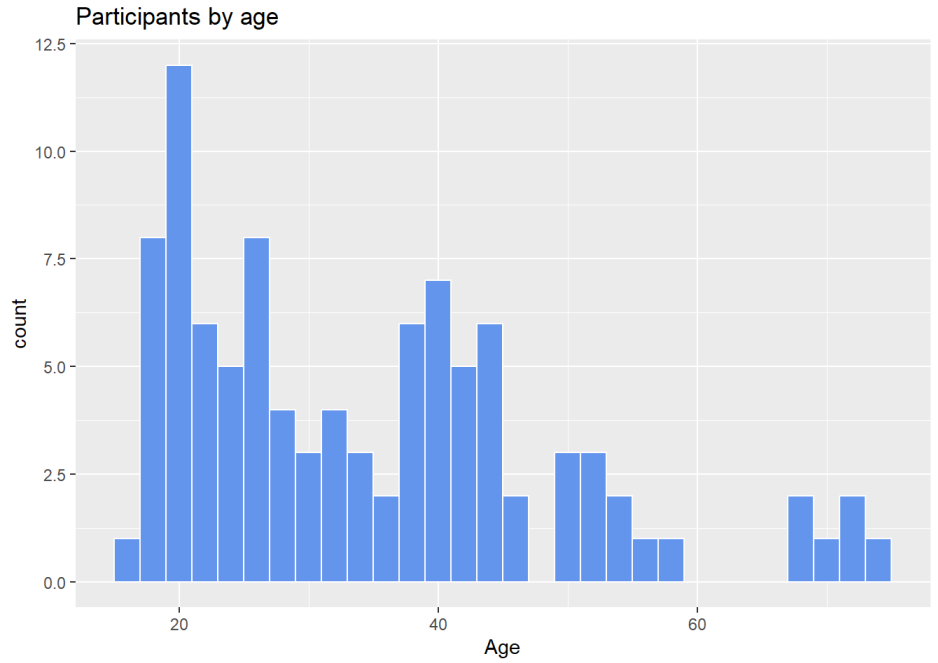


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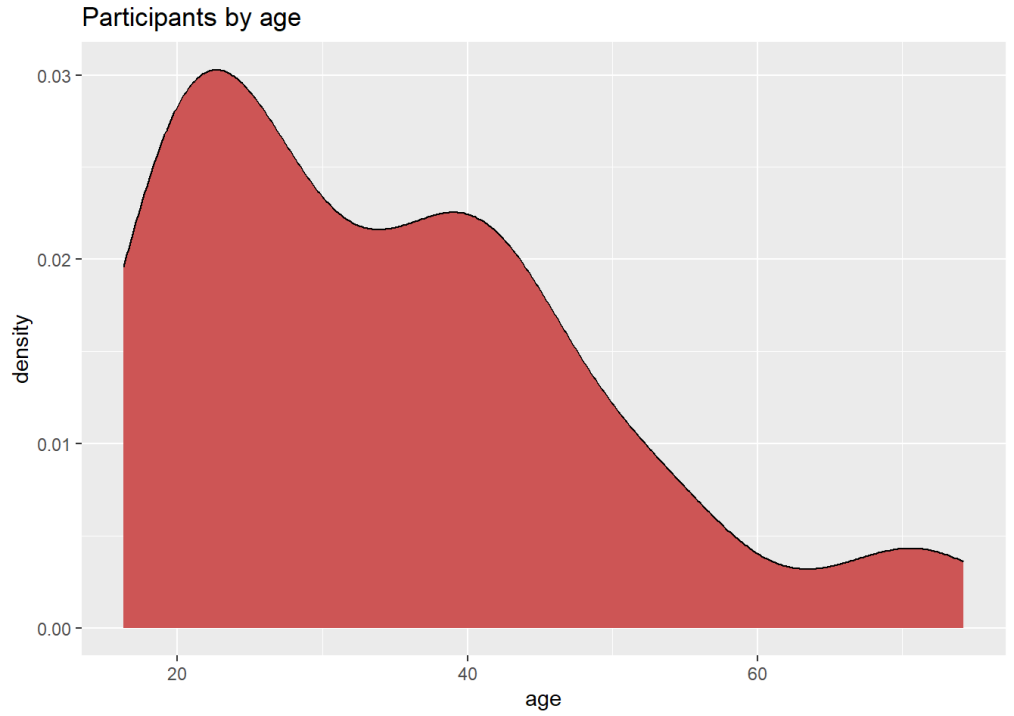
Exploratory Data Analysis

Univariate graphs for quantitative variable:-

Histogram is used to display the distribution of participants by age.



Kernel density plot is basically a smoothed histogram.

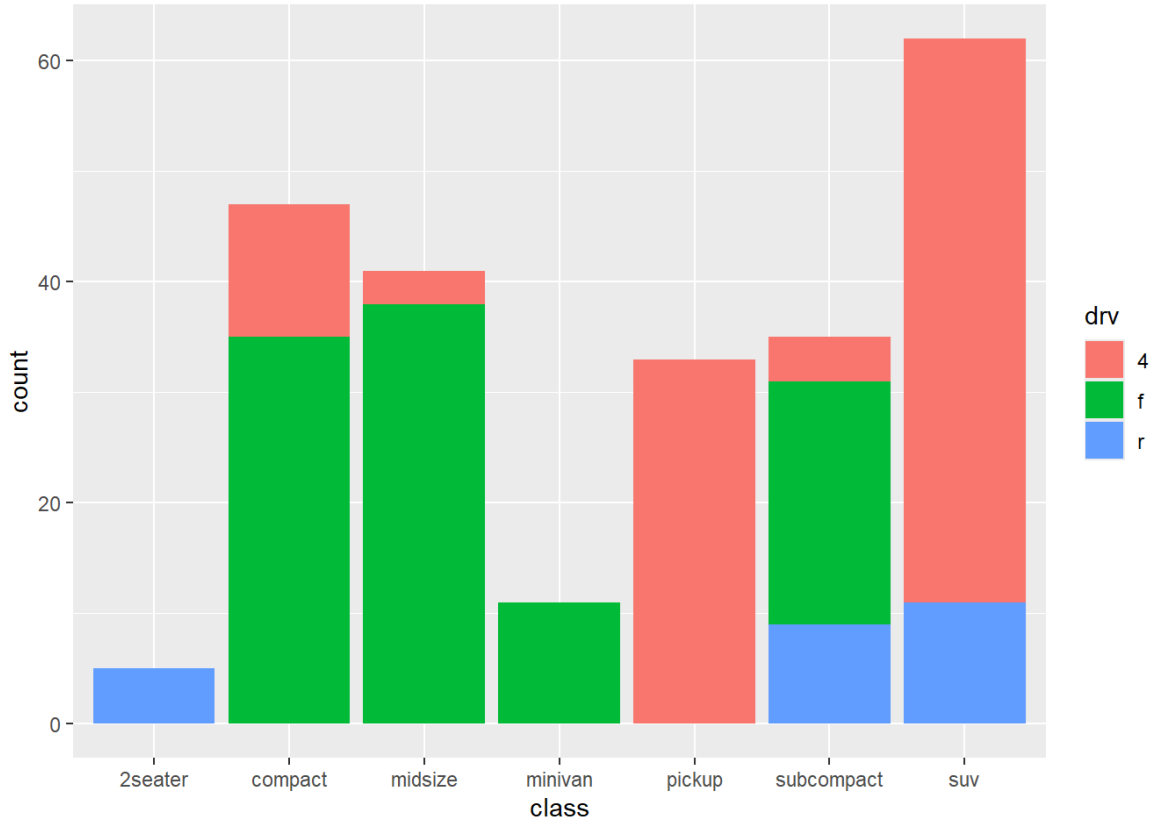


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Exploratory Data Analysis

Bivariate graphs: categorical vs categorical

Stacked bar chart is used to examine the relationship between automobile class and drive type (front-wheel, rear-wheel, or 4-wheel drive).

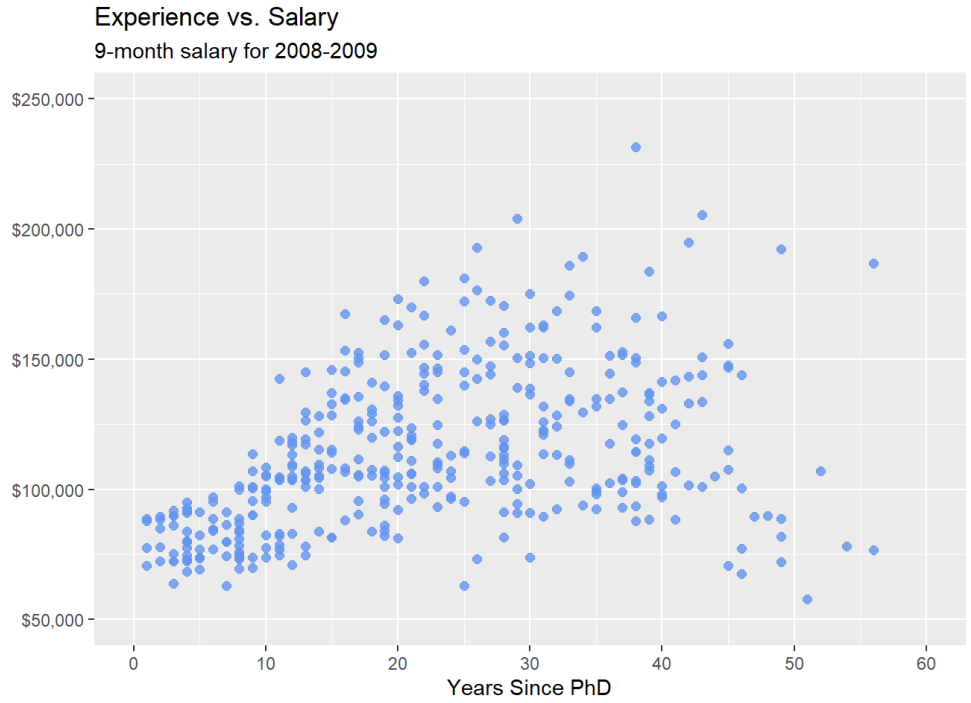


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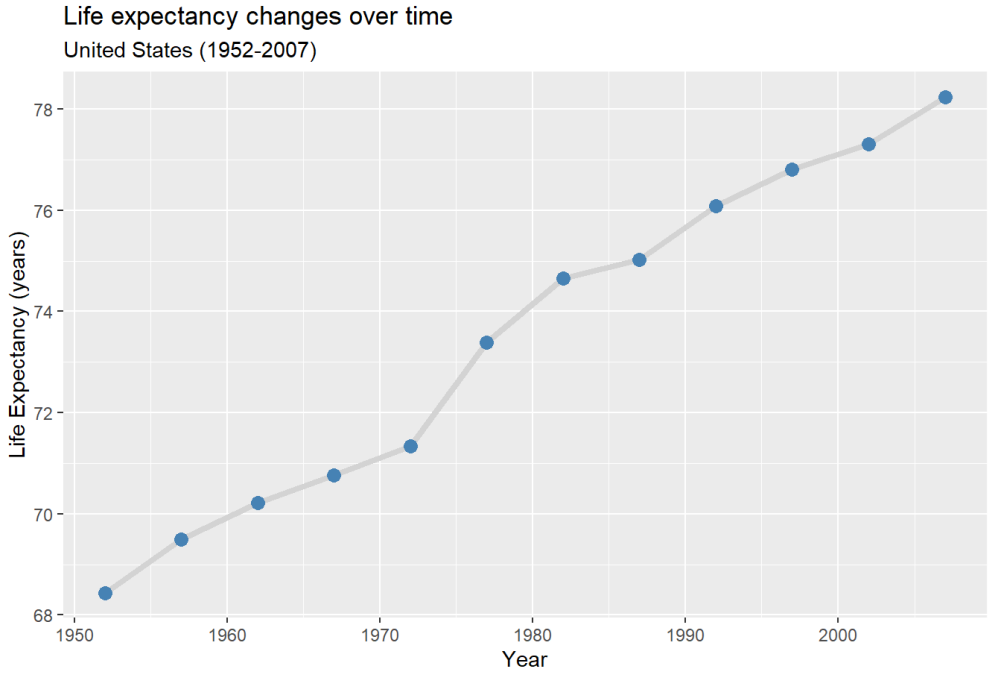
Exploratory Data Analysis

Bivariate graphs: quantitative vs quantitative

Scatterplot is used to **examine the relationship between experience (years since PhD) and academic salary for college professors**.



Line plot is used to display the relationship between time (year) and life expectancy (lifeExp) in the United States between 1952 and 2007.



Source: http://www.gapminder.org/data/

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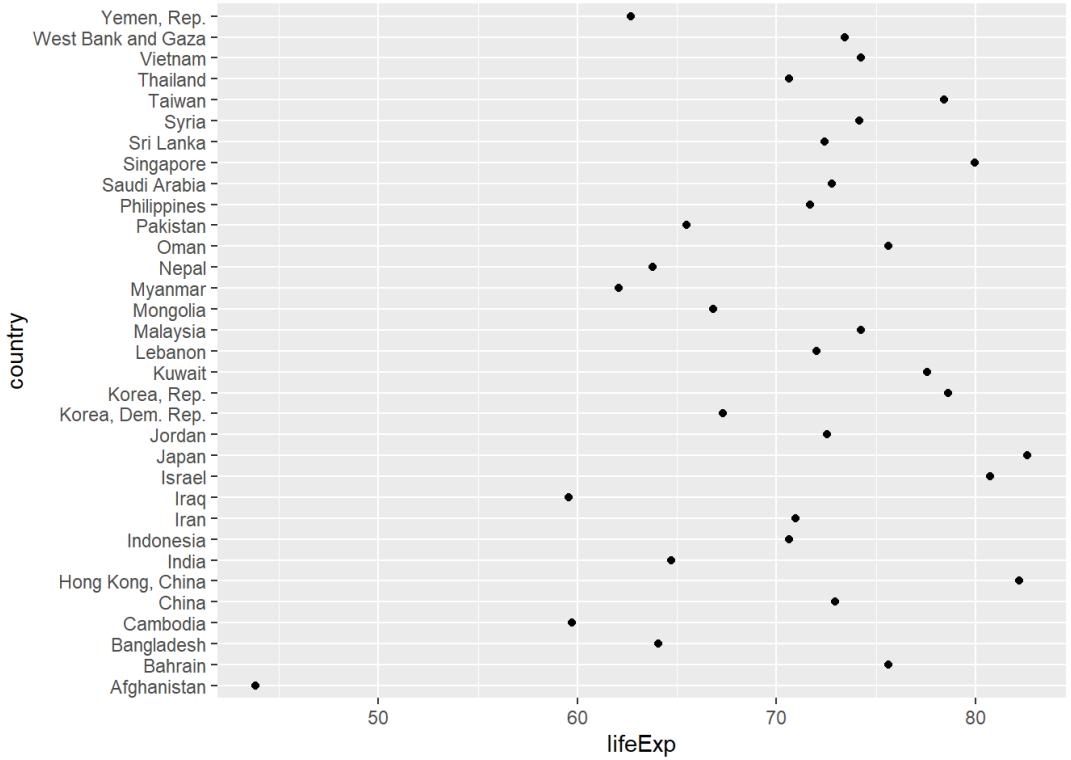
Exploratory Data Analysis

Bivariate graphs: categorical vs quantitative

Bar chart is used to display the mean salary of university professors by their academic rank.



Cleveland plot is used to compare the 2007 life expectancy for Asian country.

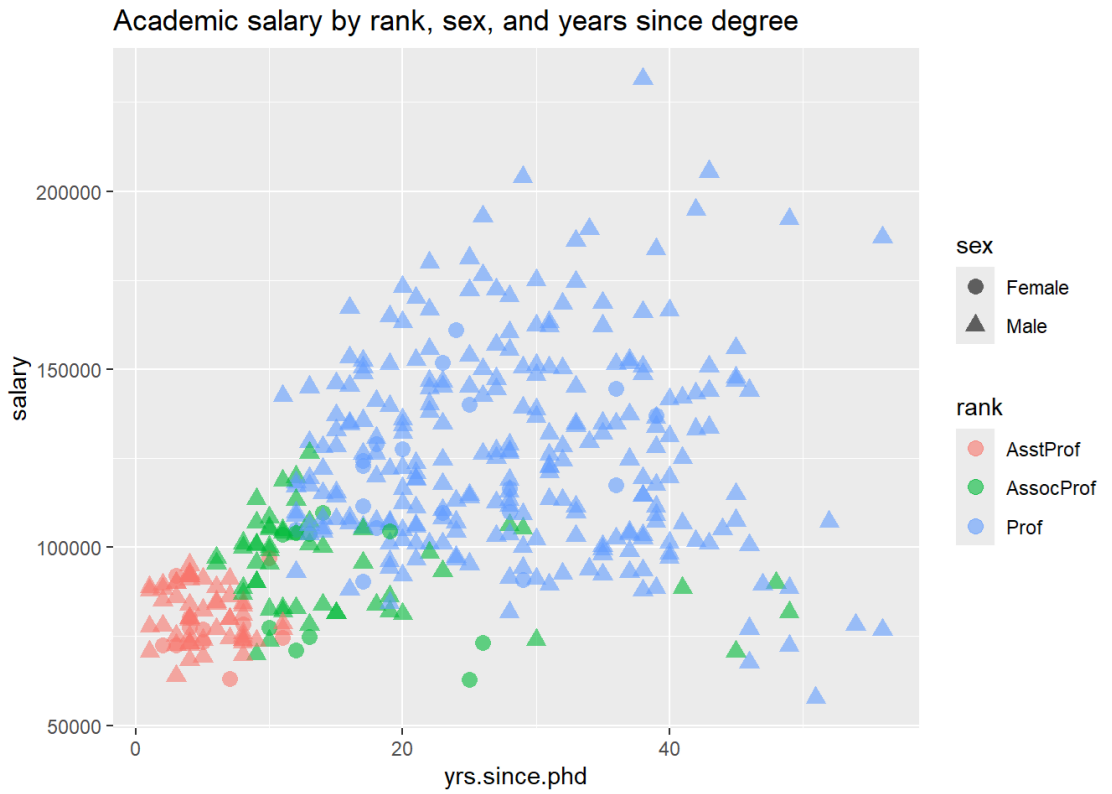


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Exploratory Data Analysis

Multivariate graphs: Grouping

In grouping, the values of the first two variables are mapped to the x and y axes. Additional variables are mapped to other visual characteristics (color, shape, line type, and transparency).

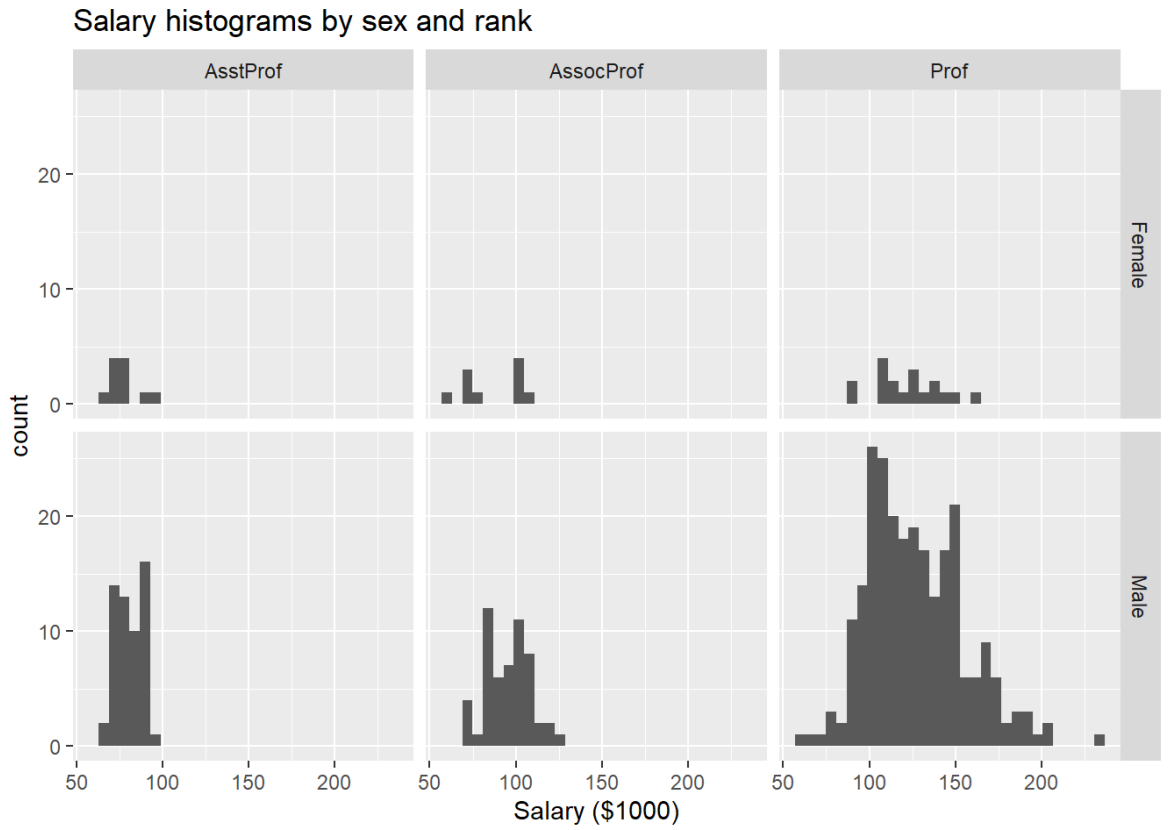


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Exploratory Data Analysis

Multivariate graphs: Faceting

Faceting consists of **several separate plots or small multiples**, one for each level of a third variable, or combination of two variables.



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Exploratory Data Analysis in Python

Step 1: Understand the Problem and the Data

Step 2: Import and Inspect the Data

Step 3: Handling Missing Values

Step 4: Explore Data Characteristics

Step 5: Perform Data Transformation

Step 6: Visualize Data Relationships

Step 7: Handling Outliers

Step 8: Discuss findings and insights

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Risks of collecting and using data

- **Privacy and security concerns**

  Medical information from the healthcare industry are sensitive and has its own regulations.

  Some medical data can be highly vulnerable to cyberattacks.

- **Misinterpretation and bias**

  Incorrect analysis or interpretation of data can influence flawed decision makings or conclusions.

- **High costs and time investment**

  Manual labelling data can be labor intensive and costly.

THANK YOU

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